

Certificate Generation System

1. Objectives

- Automate certificate generation for courses, workshops, and events.
- Support bulk certificate creation using CSV/XLSX participant data.
- Reduce manual effort and eliminate certificate formatting errors.
- Generate QR code-enabled certificates for easy verification.
- Maintain certificate generation history in a SQLite database.
- Provide an analytics dashboard for monitoring certificate activity.
- Allow management of multiple certificate templates.

2. Tools Used

Tool	Purpose
Python 3.12	Backend Development
Flask	Web Framework
SQLite	Database Management
Pandas	CSV/XLSX Processing
OpenPyXL	Excel File Handling
ReportLab	PDF Certificate Generation
Pillow	Image Processing
qrcode	QR Code Generation
Bootstrap 5	Responsive User Interface
HTML5, CSS3, JavaScript	Frontend Development
Chart.js	Analytics Dashboard
VS Code	Development Environment
Git	Version Control

3. Implementation Process

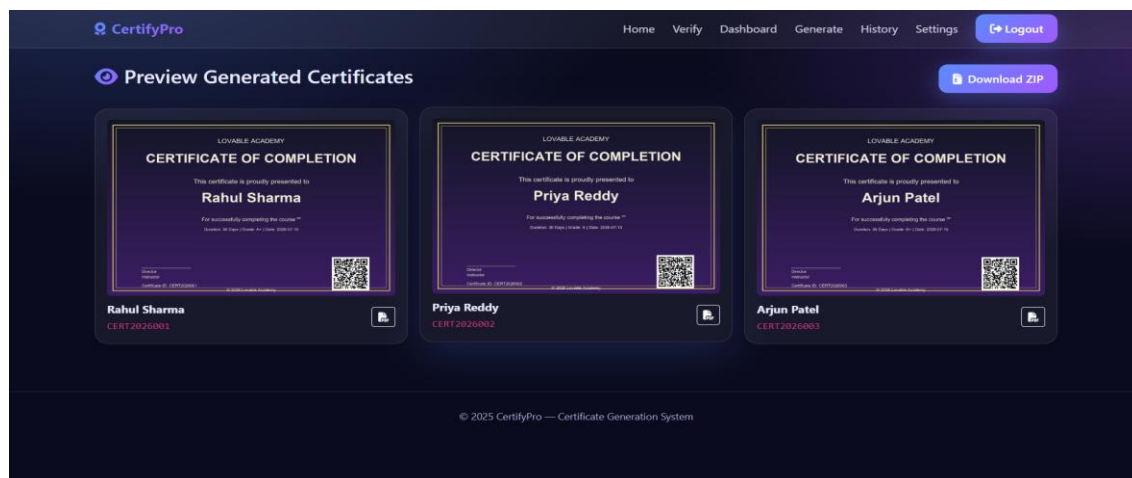
1. Administrator logs into the system.
2. Participant details are uploaded using CSV or XLSX files.
3. Uploaded data is validated for missing values, duplicate IDs, and invalid email formats.
4. An appropriate certificate template is selected.
5. QR codes are generated for each certificate.

6. Certificates are created in PDF format using ReportLab.
7. Certificate information is stored in the SQLite database.
8. Certificates can be previewed, downloaded individually, or exported as a ZIP archive.
9. Certificates can be verified using the QR code or Certificate ID.

4. Outcomes / Results

The Certificate Generation System successfully automates the process of creating and managing digital certificates. It supports bulk certificate generation, QR code-based verification, template management, history tracking, analytics, and secure administrator access. The system significantly reduces manual effort, improves accuracy, and provides a professional solution suitable for educational institutions, training organizations, and events.

Feature	Status
Login	Completed
Dashboard	Completed
CSV Upload	Completed
XLSX Upload	Completed
Certificate Generation	Completed
QR Verification	Completed
History Management	Completed
Analytics Dashboard	Completed
ZIP Export	Completed



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HomeVerifyDashboardGenerateHistorySettingsLogout

Generate Certificates

1. Upload participant data

Drag & drop CSV/XLSX here, or click to browse
sample_participants.csv

Template
Default (built-in)

Generate

2. Manage templates

Choose FileNo file chosen

Upload Template

certificate_template.png

Expected columns

Certificate ID, Participant Name, Email, Course, Organization, Date, Duration, Grade, Instructor, Remarks

Only "Participant Name" is required; missing IDs will be auto-generated.

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HomeVerifyLogin

Admin Login

Default: admin / admin123

Username

Password

Sign In

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